YoungPyo Hong

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EDUCATION

Integrated M.S./Ph.D. Program in Astronomy Sep. 2025 – present
Seoul National University, Department of Physics and Astronomy
Advisor: Prof. Myungshin Im · **M.S. expected:** Aug. 2027 · **Thesis (in progress):** Optical afterglows of gamma-ray bursts

B.S. in Physics Mar. 2020 – Feb. 2023
Yonsei University, Mirae Campus
GPA: 4.05 / 4.3 · graduated one year early
– Ranked **first in the department** at graduation · Admitted as the **top-ranked student** of the entering class

RESEARCH INTERESTS

– **Time-domain astronomy**: gamma-ray bursts, kilonovae, and gravitational-wave counterparts
– **Multi-wavelength** and **multi-messenger** follow-up of transient events
– **Machine learning** and **AI** applications in astronomy

RESEARCH EXPERIENCE

Graduate Researcher — Transient Follow-up with 7DT and RASA36 Sep. 2025 – present
Seoul National University · Advisor: Prof. Myungshin Im
– Built an **end-to-end automated rapid ToO system** for RASA36 integrating TCSpy and gpPy; **observations start within 30 s** of an alert, and the **full process** from ToO request to data reduction and transient search completes in **~40 min**
– Performed multi-band optical **SED analysis of gamma-ray burst afterglows** (e.g., GRB 251013C) with 7DT
– Designed a **near-infrared (NIR) adaptive optics (AO) system** as a graduate-course team project; produced Zemax optical designs, CAD drawings, and a technical manual, and delivered the design to an optical communications company (Spacebeam)

Post-Baccalaureate Research Intern Sep. 2024 – Aug. 2025
Seoul National University, Department of Physics and Astronomy · Advisor: Prof. Myungshin Im
– Characterized the **transmittance of 15 new medium-band 7DT filters** and validated the measurement procedure
– Built a **Slack-integrated GCN alert bot** that parses incoming GCN notices, posts summarized alerts to Slack, and automatically submits target-of-opportunity (ToO) requests

Undergraduate Research — Radio Observing Project Nov. 2021 – Feb. 2022
Gwacheon National Science Museum radio telescope
– **Proposal accepted for telescope time**: "Comparison between the distance of radio-loud quasars and their radio power"
– **Carried out on-site and remote observations**; analyzed 18 h of data (source vs. background); no significant detection

OBSERVING EXPERIENCE

RASA36 (0.36 m wide-field telescope) Sep. 2025 – present
El Sauce Observatory, Chile · remote observing
– Remote follow-up of **one GRB** and **two Einstein Probe (EP) transients** with the **automated ToO system** (upper limits)

7-Dimensional Telescope (7DT; 20-unit array, 40 medium-band filters) Sep. 2025 – present
El Sauce Observatory, Chile · remote observing
– Led ToO follow-up and GCN reports of **two GRB afterglow** detections (GRB 251013C, GRB 260223A) with medium-band SEDs; co-authored two upper-limit reports

PUBLICATIONS

Paek, G. S. H., Im, M., Jeong, M., Chang, S.-W., Hur, M. M., **Hong, Y. P.**, Kim, S., Lee, J., Lee, D., & Lee, S.-H. 2025, "GECKO Follow-up Observation of the Binary Neutron Star–Black Hole Merger Candidate S230518h," *The Astrophysical Journal*, 981, 38

Tak, D., **Hong, Y. P.**, Im, M., Maksut, Z., Kim, J. H., Chang, S.-W., Paek, G. S. H., Choi, H., & Hyun, D., "Medium-band observations of GRB 251013C with the 7-Dimensional Telescope," (in prep.)

GCN Circulars

Hong, Y. P., Im, M., Tak, D., et al. 2026, GCN Circ. 43863, "GRB 260223A: 7DT detection and Medium-band SED of Afterglow"

Hong, Y. P., Tak, D., Im, M., et al. 2025, GCN Circ. 42569, "GRB 251013C: 7DT detection and Medium-band SED of Afterglow"

Tak, D., Im, M., **Hong, Y. P.**, et al. 2025, GCN Circ. 43270, "GRB 251230A: 7DT optical upper limits"

Tak, D., Im, M., **Hong, Y. P.**, et al. 2025, GCN Circ. 43206, "GRB 251221A: 7DT optical upper limits"

CONFERENCE PRESENTATIONS

International

Cho, J., et al., incl. **Hong, Y. P.** 2022, "Performance and Application of the Radio Telescope in Gwacheon National Science Museum," *IAU General Assembly*, Busan, Korea [talk]

Domestic

Hong, Y. P., Im, M., Choi, H., Kim, S., & Paek, G. S. H. 2026, "Automated Wide-Field Rapid Follow-up System for Transient Events with TCSpy and gpPy," Korean Astronomical Society Fall Meeting, Gwacheon, Korea [talk]

Hong, Y. P., Im, M., Choi, H., Kim, S., & Paek, G. S. H. 2026, "Automated Wide-Field Rapid Follow-up System for Transient Events with TCSpy and gpPy," *Instrumentation for Astronomy and Space Science (IASS)*, Yeosu, Korea [talk]

Hong, Y. P., Tak, D., Im, M., Kim, J. H., Chang, S.-W., Paek, G. S. H., Choi, H., Hyun, D., & Lee, W.-H. 2026, "Multi-band Optical SED Analysis of GRB 251013C Afterglow with the 7-Dimensional Telescope," Korean Astronomical Society Spring Meeting, Suwon, Korea [talk]

Hong, Y. P., Im, M., Moon, H., Kim, J. H., & Choi, C. 2025, "Filter Transmittance Characterization for 15 New Medium-Band Filters of the 7-Dimensional Telescope: Measurement Procedures and Validation Results," *IASS*, Osong, Korea [poster]

SCHOOLS AND WORKSHOPS

- **Astro AI Asian Network (A3N) Summer School** (2024, 2025) — statistical modeling and deep learning for astronomy
- **Summer / Winter School on Numerical Relativity and Gravitational Waves** (2022, 2023, 2024) — GW data analysis
- **Korea Institute for Advanced Study (KIAS) Summer School on Extragalactic Astronomy and Cosmology** (2023, 2024) — hydrodynamic and N-body simulations, SED fitting, CMB data processing
- **ALMA Summer School** (2026) — ALMA data reduction and polarization analysis
- **ALMA Proposal Preparation Workshop** (2025) — training in ALMA proposal development and practice proposal writing
- **K-GMT Summer School** (2023) — Gemini/GNIRS proposal: "Further insight on galaxy rotation curves at high redshift with GNIRS"
- **Radio Summer School** (2022) — receiver systems, interferometry, AGN, SKA
- **Workshops:** SKA-Korea (2024, 2025), Korea Rubin Science (2025), LSST with AI (2025), Woojeon (2025, 2026), 10th Anniversary of the Discovery of Gravitational Waves (2025), Extragalactic Transients (2026)

AWARDS AND HONORS

- **Next Generation of Researchers in Basic Sciences Fellowship**, Seoul National University (university-wide competitive selection; KRW 13M (~USD 9k)/year) — Feb. 2026
- **Yonsei Vision Scholarship**, Yonsei University (academic excellence, previous semester) — Sep. 2022
- **National Program of Excellence in Software Scholarship**, Yonsei University — Mar. 2021, Mar. 2022
- **Semester Honors and Truth Scholarship**, Yonsei University (Fall 2021; partial tuition, academic excellence) — Feb. 2022
- **Yonsei Special Scholarship** (full tuition; admission with highest distinction) — Mar. 2020

TEACHING AND OUTREACH

Teaching Assistant, Seoul National University — Sep. 2025 – present
Man and the Universe (Fall 2025, Fall 2026); Astronomical Observation and Laboratory II (Spring 2026)

Astronomy Communicator, Level II, Korean Amateur Astronomical Society — 2022
Trained in public outreach and in presenting astronomical concepts to non-expert audiences.

Physics Tutor for First-Year Undergraduates, Yonsei University — Mar. – Jun. 2022

Korean University Knowledge Mentoring Program — Apr. 2021 – Jan. 2022
Academic tutoring and career counseling for elementary, middle, and high school students.

MILITARY SERVICE

Republic of Korea Army — mandatory military service — Apr. 2023 – Jul. 2024